UNIT – III

MECHANICS OF HARD AND SOFT TISSUES

Introduction

- Bone is an important and dynamic living tissue.
- Its mechanical functions are to support and protect other body tissues and to act as a system of rigid levers that can be manipulated by the attached muscles.
- Bone’s strength and resistance to fracture depend on its material composition and organizational structure.
- Minerals contribute to a bone’s hardness and compressive strength, and collagen provides its flexibility and tensile strength.

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- The material constituents and structural organization of bone influence the ways in which bone responds to mechanical loading.
- The composition and structure of bone yield a material that is strong for its relatively light weight.

Material Constituents

- The major building blocks of bone are calcium carbonate, calcium phosphate, collagen, and water.
- The relative percentages of these materials vary with the age and health of the bone.
- Calcium carbonate and calcium phosphate generally constitute approximately 60–70% of dry bone weight.
- These minerals give bone its stiffness and are the primary determiners of its compressive strength.
- Other minerals, including magnesium, sodium, and fluoride, also have vital structural and metabolic roles in bone growth and development.

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- Collagen is a protein that provides bone with flexibility and contributes to its tensile strength.
- The water content of bone makes up approximately 25–30% of total bone weight.
- The water present in bone tissue is an important contributor to bone strength.
- For this reason, scientists and engineers studying the material properties of different types of bone tissue must ensure that the bone specimens they are testing do not become dehydrated.

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- The flow of water through bones also carries nutrients to and waste products away from the living bone cells within the mineralized matrix.
- In addition, water transports mineral ions to and from bone for storage and subsequent use by the body tissues when needed.

- Afterload is defined as the ventricular wall stress or tension that develops during systolic contraction and ejection of blood into the aorta.
- Actin molecules are present in all muscles, leukocytes, red blood cells, endothelial cells, and many other cells.
- Actin monomers are globular.
- They polymerize into filaments.
- Actin filaments labelled with phalloidin tetramethyl-rhodamine are stable
- A ligament is an elastic band of tissue that connects bone to bone and provides stability to the joint.
- Cartilage is soft, gel-like padding between bones that protects joints and facilitates movement.
- Tendons and ligaments do not have the ability to contract like muscle tissue, but they are slightly extensible. These tissues are elastic and will return to their original length after being stretched, unless they are stretched beyond their elastic limits.

Structural Organization

- The relative percentage of bone mineralization varies not only with the age of the individual but also with the specific bone in the body.
- Some bones are more porous than others.
- The more porous the bone, the smaller the proportion of calcium phosphate and calcium carbonate, and the greater the proportion of nonmineralized tissue.

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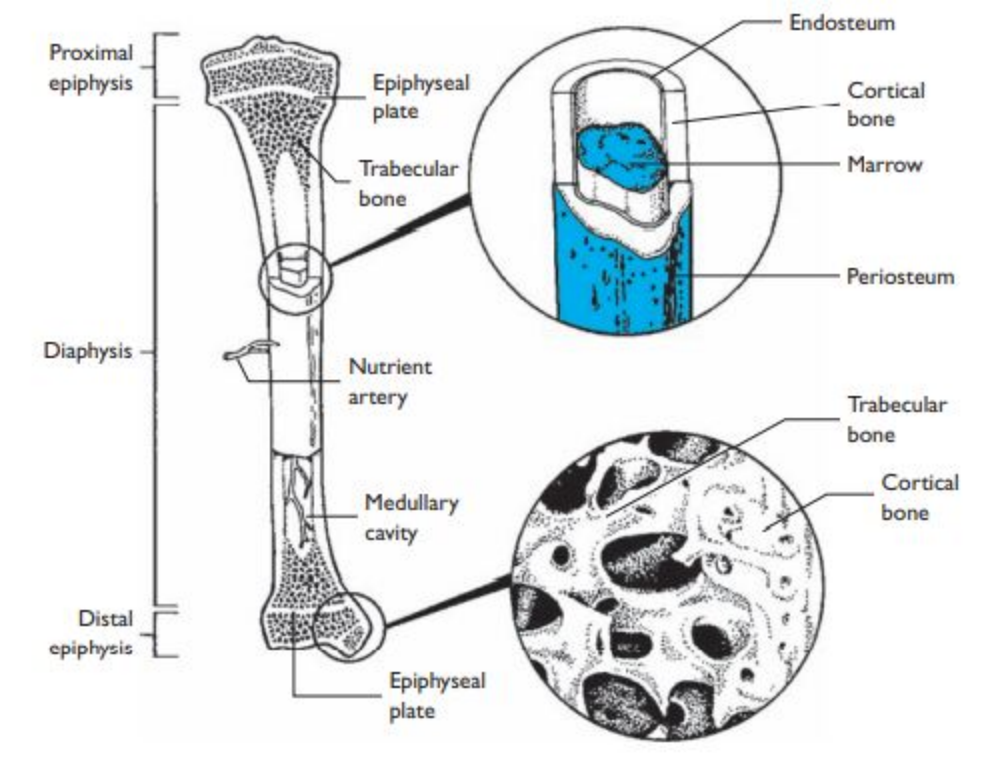
- Bone tissue has been classified into two categories based on porosity.
- If the porosity is low, with 5–30% of bone volume occupied by nonmineralized tissue, the tissue is termed cortical bone.
- Bone tissue with a relatively high porosity, with 30% to greater than 90% of bone volume occupied by nonmineralized tissue, is known as spongy, cancellous, or trabecular bone.

Structure of Cortical and trabecular bone



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- Trabecular bone has a honeycomb structure with mineralized vertical and horizontal bars, called trabeculae, forming cells filled with marrow and fat.
- The porosity of bone is of interest because it directly affects the mechanical characteristics of the tissue.
- With its higher mineral content, cortical bone is stiffer, so that it can withstand greater stress, but less strain or relative deformation, than trabecular bone.
- Because trabecular bone is spongier than cortical bone, it can undergo more strain before fracturing.

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- The function of a given bone determines its structure.
- The shafts of the long bones are composed of strong cortical bone.
- The relatively high trabecular bone content of the vertebrae contributes to their shock absorbing capability.
- Both cortical and trabecular bone are anisotropic; that is, they exhibit different strength and stiffness in response to forces applied from different directions.
- Bone is strongest in resisting compressive stress and weakest in resisting shear stress

Mechanical Properties of Bone

- The mechanical characteristics of the bone depend on the material from which the bone is built as well as on its shape, size and structure.
- There are two kinds of mechanical and structural strength.
- Both types have a significant dynamic component: each of the strengths is constantly tuned to the type of mechanical stimulus that affects the locomotor system.
- Decreased physical activity can cause pathological changes in mechanical strength of bone tissue.

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- Mechanical parameters depend on the content of organic and inorganic materials.
- Inorganic materials are responsible for giving the bones elastic properties, are related to the activity of cells which in turn depends on the correct blood supply of the bone tissue.
- The occurrence of ischemia as well as deprivation induces dramatic changes in mechanical properties.

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- The forces to which the bone is usually exposed include compression, tension, torsion, bending and shear stress.
- In case of compression, the bone is exposed to two forces - the compressive force itself and the reaction force, in accordance with Newton's third law of dynamics.
- In pure compression, the compressive and reaction force vectors are directly opposite to each other, creating uniaxial compression

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- The compressive strength of the bone tissue ranges from 12.56 to 16.87 kg / mm2 in cross section and is a value that approaches the tensile strength of the bone.
- The mechanical strength of individual bones for compression forces varies.
- The patella is broken down by the force of 192 kg, the humerus 600 kg, the femur 756 kg, and the tibiae 450 kg.
- These values reflect the forces usually causing bone fracture, it is worth remembering that fractures are rarely the result of compression itself.

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- Tensile strength of the dry bone is 10 kg / mm2 cross-sectional area, while for newly formed compact bone the tensile strength is 12.41 kg / mm2 .
- For stretching, cortical bone is characterized by greater stress, with less deformation compared to cancellous bone.
- In the range of elastic deformations, the length of the cancellous bone may increase by 50% at 1.5 to 2% of cortical bone growth.

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- The mechanical properties of bone tissue are also affected by many diseases, the most common being osteoporosis.
- It is a systemic disease characterized by bone mass loss and microarchitectural disorders, leading to the weakening of the bone structure.
- Osteoporosis is the reduction of bone mass and density and the deterioration of bone quality.

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- This reduction is a cause of increased resorption, which is not accompanied by proper deposition of the new bone, initially it occurs in the cancellous bone, and then the changes also occur within compact bones.
- After adolescence, only a small increase in bone mass is observed, and after the age of 30 years, the bone density decreases by approximately 0.7% per year.

Anisotrophy

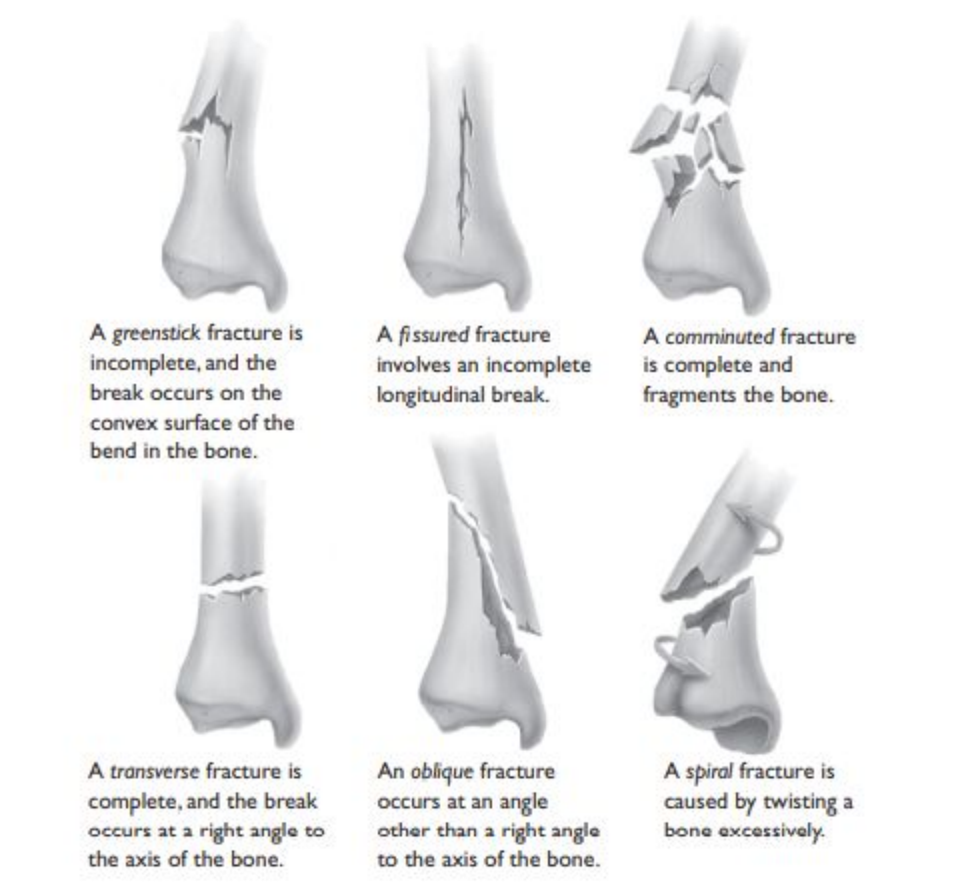
- Bone is said to be anisotropic i.e., material properties are different in different directions.
- Bone is composed of mainly collage and other minerals (hydroxyapatite or calcium).
- Collagen has high tensile strength and low compressive strength.
- Hydroxyapatite is stiff, brittle and has high compressive strength.
- Anisotropic material can resist different types of forces and is strongest in compression, weakest in shear and intermediate in tension.

- Because of the important mechanical functions performed by bone, bone health is an important part of general health.
- Bone health can be impaired by injuries and pathologies.

Fracture

- A fracture is a disruption in the continuity of a bone.
- The nature of a fracture depends on the direction, magnitude, loading rate, and duration of the mechanical load sustained, as well as the health and maturity of the bone at the time of injury.
- Fractures are classified as –
  - simple when the bone ends remain within the surrounding soft tissues
  - compound when one or both bone ends protrude from the skin.

Types of Fractures



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- When the loading rate is rapid, a fracture is more likely to be comminuted, containing multiple fragments.
- Avulsions are fractures caused by tensile loading in which a tendon or ligament pulls a small chip of bone away from the rest of the bone.
- Explosive throwing and jumping movements may result in avulsion fractures of the medial epicondyle of the humerus and the calcaneus.

Spiral Fracture

- Excessive bending and torsional loads can produce spiral fractures of the long bones.
- The simultaneous application of forces from opposite directions at different points along a structure such as a long bone generates a torque known as a bending moment, which can cause bending and ultimately fracture of the bone.
- A bending moment is created on a football player’s leg when the foot is anchored to the ground and tacklers apply forces at different points on the leg in opposite directions.
- When bending is present, the structure is loaded in tension on one side and in compression on the opposite side.

Greenstick Fracture

A **greenstick fracture** is an **incomplete fracture** in which one side of the bone breaks while the other side bends.

- It is commonly seen in **children** because their bones are softer and more flexible.

Causes

- Fall on an outstretched hand
- Sports injury
- Direct trauma

Clinical Features

- Pain
- Swelling
- Tenderness
- Deformity (mild bending)

Treatment

- Immobilization with cast or splint
- Usually heals faster than complete fracture
- Surgery rarely required

Oblique Fracture

An **oblique fracture** is a type of bone fracture in which the break occurs at an **angle (diagonal line)** across the bone shaft.

It is commonly seen in **children** because their bones are softer and more flexible.

Causes

- Direct blow at an angle
- Twisting injury
- Fall or accident
- Sports trauma

Characteristics

- Fracture line runs diagonally
- Bone is completely broken
- May cause displacement of bone fragments

Treatment

- Immobilization with cast
- Traction
- Surgery with plates or screws (if displaced)

Impacted fracture

- Since bone is stronger in resisting compression than in resisting tension and shear, acute compression fractures of bone are rare.
- However, under combined loading, a fracture resulting from a torsional load may also be impacted by the presence of a compressive load.
- An impacted fracture is one in which the opposite sides of the fracture are compressed together.
- Fractures that result in depression of bone fragments into the underlying tissues are termed depressed.

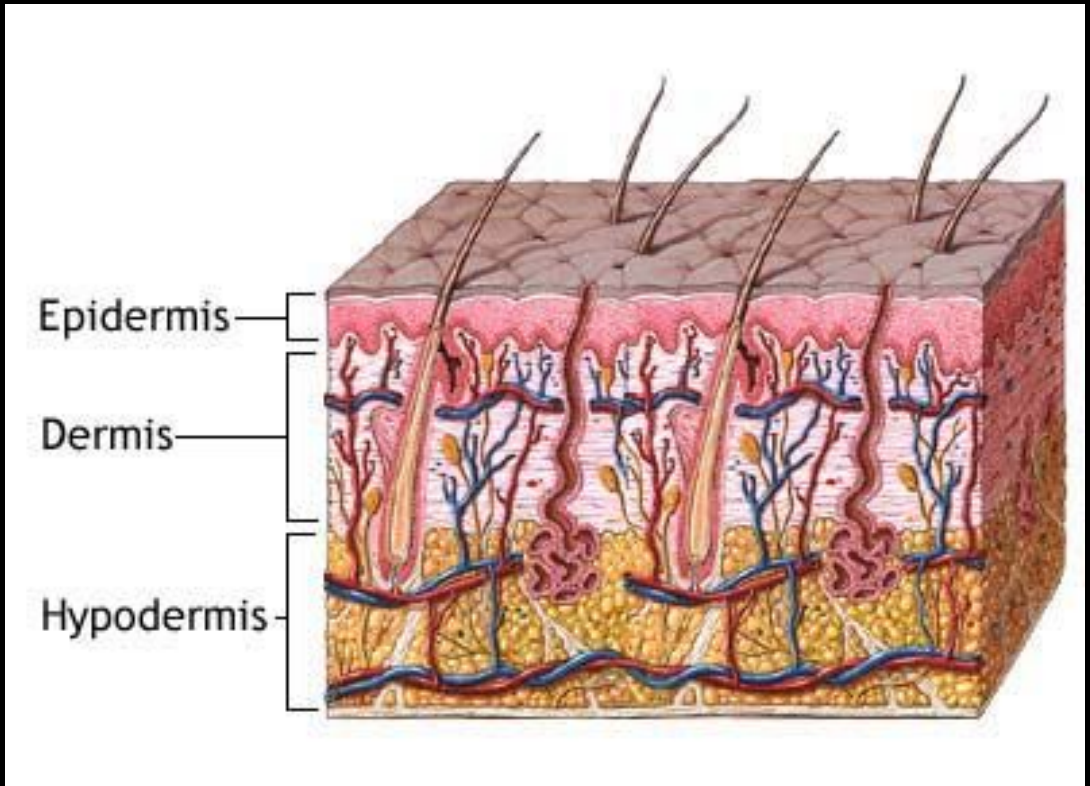
Stress fracture

- Stress fractures, also known as fatigue fractures, result from low magnitude forces sustained on a repeated basis.
- Any increase in the magnitude or frequency of bone loading produces a stress reaction, which may involve microdamage.
- Bone responds to microdamage by remodeling: First, osteoclasts resorb the damaged tissue; then, osteoblasts deposit new bone at the site.
- When there is not time for the repair process to complete itself before additional microdamage occurs, the condition can progress to a stress fracture.
- Stress fractures begin as a small disruption in the continuity of the outer layers of cortical bone but can worsen over time, eventually resulting in complete cortical fracture.

The skin

- The skin is the largest organ in the human body.
- It consists of about ten percent of our body mass.
- The skin is composed of three distinct layers: the epidermis, the dermis, and the subcutaneous fat, sometimes called the hypodermis.

Layers of Skin



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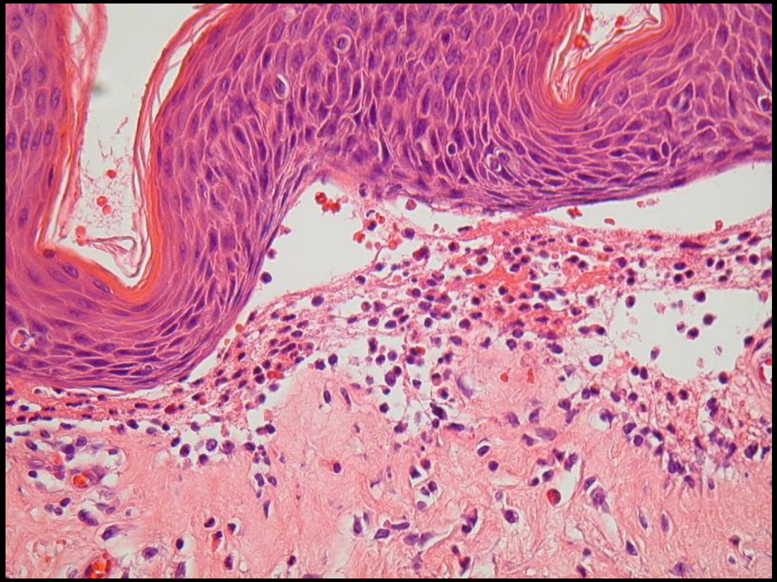
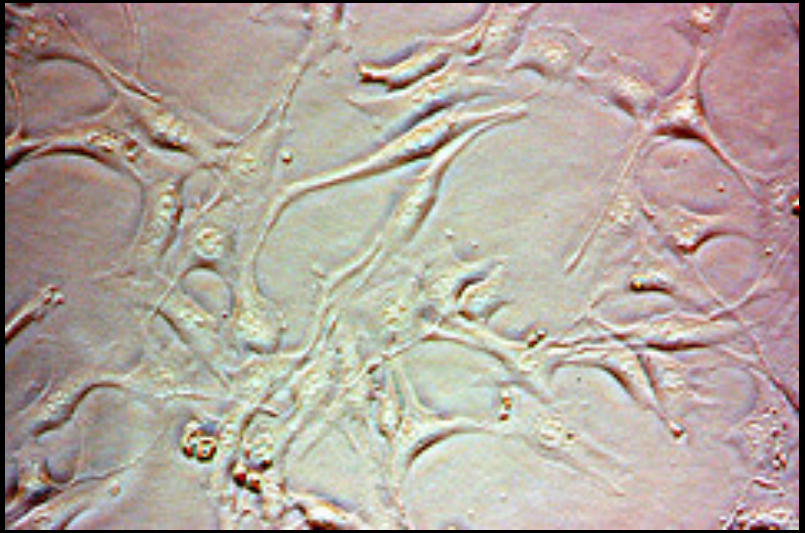
- The epidermis is the outermost skin layer and consists of keratinocytes, rapidly dividing stem cells, that help generate epidermal cells.
- The second layer, the dermis, consists of collagen fibers in a gel like state and also fibroblasts. It helps binds the epidermis so it conforms to the shape of the body.
- The deepest skin layer, the subcutaneous layer, consists mainly of fatty tissue.

Keratinocytes and Fibroblasts

Keratinocytes





Fibroblasts

Functions of Skin

- A main function of the external skin layer is to provide a tough barrier covering the entire body.
- It provides defense from radiation, disease, gases, chemicals, and many other destructive forces.
- However, if the skin is damaged it can lead to difficult complications.

Mechanical Properties of Skin

Depends on the nature and organization of:

- Dermal collagen and elastic fibers network
- Water, proteins and macromolecule embedded in the extracellular matrix
- with less contribution by epidermis and stratum corneum

Collagen Molecules

- 300 nm long and 1.5 nm in diameter
- Tropocollagen triple helix- consist of three polypeptide strands
- Quaternary structure (stabilized by hydrogen bonds)
- 29 types of collagen
- E along fiber ~ 1000 Mpa
- UTS ~ 50-100 MPa

Types of Collagen

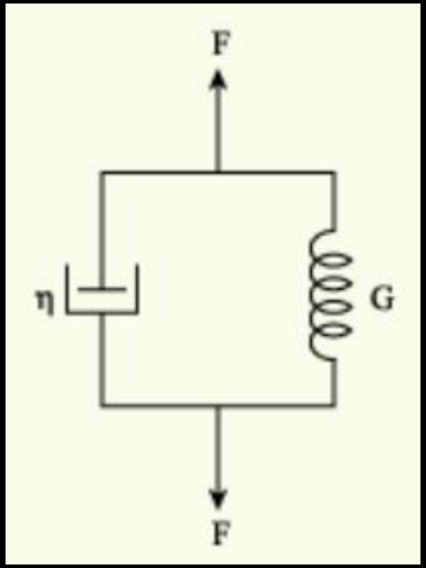
- Collagen I: skin, tendon, vascular, ligature, organs, bone (main component of bone)
- Collagen II: cartilage (main component of cartilage)
- Collagen III: reticulate (main component of reticular fibers), commonly found alongside type I.
- Collagen IV: basis of cell basement membranes
- Collagen V: Cells surfaces, hair and placenta

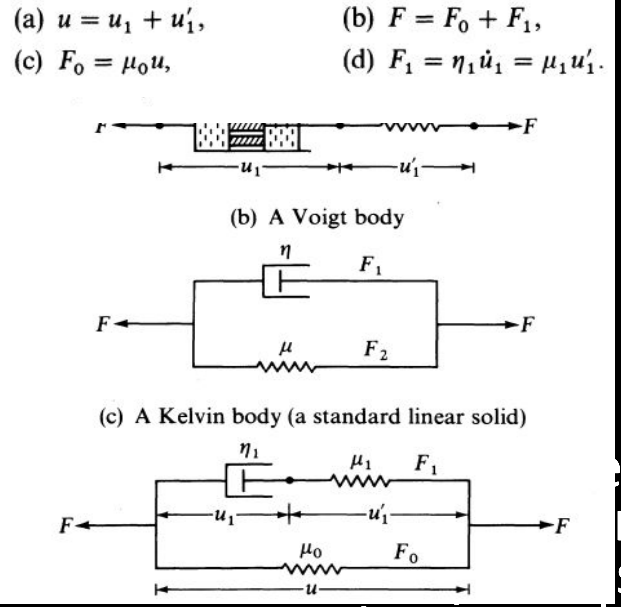
Elastin

- Protein fibrillin and amino acids (glycine, valine, alanine, and proline)
- Providing elasticity- tissue are able to retract back to its shape after deformation
- Location- blood vessels (Windkessel effect), lungs, skin, bladder and elastic cartilage.
- E ~ 0.6 MPa

Viscoelasticity

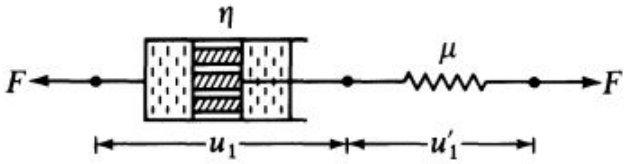
- Skin exhibit both viscous and elastic characteristics when undergoing deformation



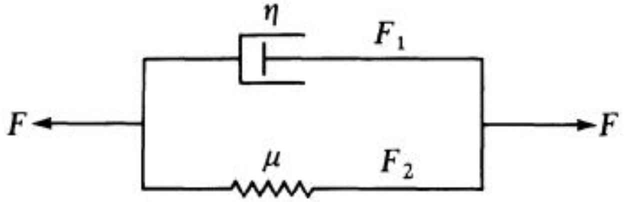


is suddenly strained and then the strain is maintained constant corresponding stresses induced in the body decrease with time. na is called stress relaxation, or relaxation for short. If the body is sed and then the stress is maintained constant afterward, the body continues to deform, and the phenomenon is called creep. If the body is subjected to a cyclic loading, the stress-strain relationship in the loading process is usually somewhat different from that in the unloading process, and the phenomenon is called hysteresis. The features of hysteresis, relaxation, and creep are found in many materials. Collectively, they are called features of viscoelasticity. Mechanical models are often used to discuss the viscoelastic behavior of materials. In Fig: 1 are shown three mechanical models of material behavior, namely, the Maxwell model, the Voigt model, and the Kelvin model (also called the standard linear solid), all of which are composed of combinations of linear springs with spring constant J1 and dash pots with coefficient of viscosity µf.

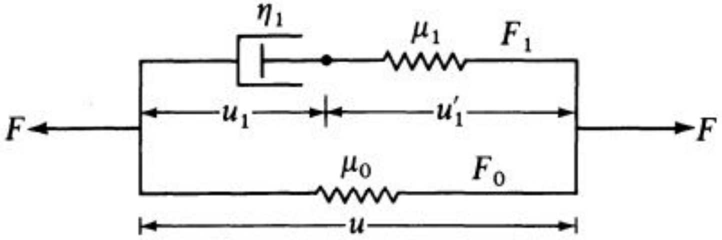
(a) A Maxwell body



(b) A Voigt body



(c) A Kelvin body (a standard linear solid)



$$\dot{u} = \frac{\dot{F}}{\mu} + \frac{F}{\eta} \qquad \text{(Maxwell model).}$$

- The velocity of the spring extension is F/μ if we denote a differentiation with respect to time by a dot. The total velocity is the sum of these two:

$$\dot{u} = \frac{\dot{F}}{\mu} + \frac{F}{\eta} \qquad \text{(Maxwell model).}$$

- For the Voigt model, the spring and the dashpot have the same displacement. If the displacement is u, the velocity is U, and the spring and dash pot will produce forces μU and nU, respectively. The total force F is therefore

$$F = \mu u + \eta \dot{u} \qquad \text{(Voigt model).}$$

- For the Kelvin model (or standard linear model), let us break down the displacement u into U1 of the dashpot and u~ for the spring, whereas the total force F is the sum of the force Fo from the spring and FI from the Maxwell element. Thus

$$\text{(a)}\ u = u_1 + u_1', \qquad \text{(b)}\ F = F_0 + F_1,$$
$$\text{(c)}\ F_0 = \mu_0 u, \qquad \text{(d)}\ F_1 = \eta_1 \dot{u}_1 = \mu_1 u_1'.$$

Viscoelastic properties of skin

- Mechanical behavior of skin and tendon are different.
- This is due to differences in collagen types self-assembly, i.e. tilt angle of collagens (orientation), fiber length, volume fraction of the Fibers, collagen molecular stretching.

STRUCTURAL ORGANIZATION OF SKELETAL MUSCLE

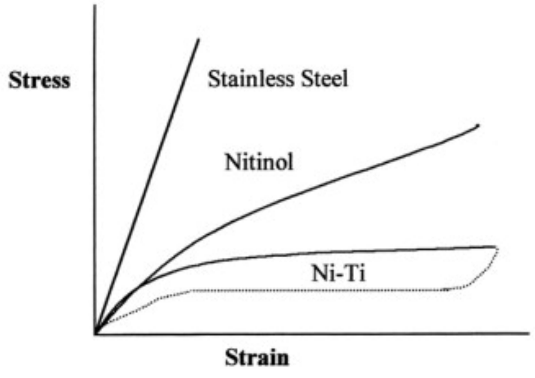
- There are approximately 434 muscles in the human body, making up 40–45% of the body weight of most adults.
- Muscles are distributed in pairs on the right and left sides of the body.
- About 75 muscle pairs are responsible for body movements and posture, with the remainder involved in activities such as eye control and swallowing.
- When tension is developed in a muscle, biomechanical considerations such as the magnitude of the force generated, the speed with which the force is developed, and the length of time that the force may be maintained are affected by the particular anatomical and physiological characteristics of the muscle.

SKELETAL MUSCLE FUNCTION

- When an activated muscle develops tension, the amount of tension present is constant throughout the length of the muscle, as well as in the tendons, and at the sites of the musculotendinous attachments to bone.
- The tensile force developed by the muscle pulls on the attached bones and creates torque at the joints crossed by the muscle.
- In keeping with the laws of vector addition, the net torque present at a joint determines the direction of any resulting movement.
- The weight of the attached body segment, external forces acting on the body, and tension in any muscle crossing a joint can all generate torques at that joint.

Tendons

- Tendons, which connect muscles to bones, and ligaments, which connect bones to other bones, are passive tissues composed primarily of collagen and elastic fibers.
- Tendons and ligaments do not have the ability to contract like muscle tissue, but they are slightly extensible. These tissues are elastic and will return to their original length after being stretched, unless they are stretched beyond their elastic limits.



- **Pseudoelasticity**, sometimes called **superelasticity**, is an elastic (reversible) response to an applied stress, caused by a phase transformation between the austenitic and martensitic phases of a crystal. It is exhibited in shape-memory alloys
- It's the property of restoring the original shape after plastically deformed. This involves the martensite to austenite. The transformation is mainly due to heating
- It's the ability of Shape memory alloy to recover large stress- strain hysteresis due to mechanical loading and unloading
- It is an elastic response to an applied stress, caused by the phase transformation between the austinite and martensite
- In bone clamping and braces this method is widely used.